

Zhian Jia^{*}

PERSONAL DATA

NAME: Zhian Jia (Surname: Jia, '贾'; Given name: Zhian '治安')
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ZHIHU: <https://www.zhihu.com/people/polyidiot>

ACADEMIC EXPERIENCE

SEP 2021 - PRESENT | Centre for Quantum Technologies, **National University of Singapore, Singapore**
Supervisor: Dagomir Kaszlikowski
Research Fellow

NOV 2018 - DEC 2019 | Microsoft Station Q, Department of Mathematics, **University of California, Santa Barbara, California, United States**
Supervisor: Zhenghan Wang
Visiting scholar program

SEP 2017 - AUG 2018 | Yau Mathematical Sciences Center, Department of Mathematical Sciences, **Tsinghua University, Beijing, China**
Supervisor: Liang Kong
Visiting Ph.D.

EDUCATION

SEP 2015 - JUN 2021 | CAS Key Laboratory of Quantum Information, **University of Science and Technology of China, Hefei, China**
Supervisor: Guang-Can Guo, Yu-Chun Wu
PhD degree in Physics
Thesis: Classification, criteria and properties of quantum correlations and their applications in quantum many-body systems

SEP 2011 - JUN 2015 | Institute of Super-microstructure and Ultrafast Process in Advanced Materials, School of Physics and Electronics, **Central South University, Changsha, China**
B.S. in Applied Physics
Thesis: Impurity effect of vacancy in two-dimensional crystals and the related applications in quantum Hall effect

SELECTED AWARDS

2017 Guorui scholarship for graduate students
2016 National scholarship for graduate students
2014 College scholarships of Physics and Electronics, Central South University
2010 *The Second Prize* of The 27th national physics olympiad, 2010.

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- **Z. Jia**, Dagomir Kaszlikowski, Sheng Tan, Boundary and domain wall theories of 2d generalized quantum double model, JHEP07(2023)160, [arXiv:2207.03970]
- **Z. Jia**, Sheng Tan, Dagomir Kaszlikowski, Liang Chang, On weak Hopf symmetry and weak Hopf quantum double model, Communications in Mathematical Physics 402, 3045–3107 (2023), [arXiv:2302.08131]
- Xiangjing Liu, **Z. Jia**, Yixian Qiu, Fei Li, Oscar Dahlsten, Unification of spatiotemporal quantum formalisms: mapping between process and pseudo-density matrices via multiple-time states, New J. Phys. 26 033008 (2024), [arXiv:2306.05958]
- **Z. Jia**, Dagomir Kaszlikowski, The spatiotemporal doubled density operator: a unified framework for analyzing spatial and temporal quantum processes, [arXiv:2305.15649]
- **Z. Jia**, Minjeong Song, Dagomir Kaszlikowski, Quantum space-time marginal problem: global causal structure from local causal information, New J. Phys. 25 123038 (2023), [arXiv:2303.12819]
- Lu Wei, **Z. Jia**, Dagomir Kaszlikowski, Sheng Tan, Antilinear superoperator and quantum geometric invariance for higher-dimensional quantum systems, [arXiv:2202.10989]
- **Z. Jia**, Dagomir Kaszlikowski, Sheng Tan, Electric-magnetic duality of $\mathbb{Z}_2$ symmetry enriched Abelian lattice gauge theory, J. Phys. A: Math. Theor. 57 255203 (2024) [arXiv:2201.12361]
- Huan Cao, Ning-ning Wang, **Z. Jia**, Chao Zhang, Yu Guo, Bi-Heng Liu, Yun-Feng Huang, Chuan-Feng Li, Guang-Can Guo, Quantum simulation of indefinite causal order induced quantum refrigeration, Phys. Rev. Research 4, L032029 (2022), [arXiv:2101.07979]
- **Z. Jia**, Lu Wei, Yu-Chun Wu, Guang-Can Guo, Quantum Advantages of Communication Complexity from Bell Nonlocality, Entropy 23 (6), 744 (2021)
- **Z. Jia**, Rui Zhai, Shang Yu, Yu-Chun Wu, and Guang-Can Guo, Hierarchy of Genuine Multipartite Quantum Correlations, Quantum Inf Process 19, 419 (2020)
- Yu Meng, Shang Yu, **Z. Jia**, Yi-Tao Wang, Zhi-Jin Ke, Wei Liu, Zhi-Peng Li, Yuan-Ze Yang, Hang Wang, Yu-Chun Wu, Jian-Shun Tang, Chuan-Feng Li, Guang-Can Guo, Environment-induced sudden change of coherence in quantum systems, Phys. Rev. A 102, 042415 (2020)
- **Z. Jia**, Lu Wei, Yu-Chun Wu, Guang-Can Guo, Guo-Ping Guo, Entanglement Area Law for Shallow and Deep Quantum Neural Network States, New J. Phys. 22 053022 (2020)
- **Z. Jia**, Biao Yi, Rui Zhai, Yu-Chun Wu, Guang-Can Guo and Guo-Ping Guo, Quantum Neural Network States: A Brief Review of Methods and Applications, Adv. Quantum Technol. 2019, 1800077
- **Z. Jia**, Yuan-Hang Zhang, Yu-Chun Wu, Liang Kong, Guang-Can Guo, and Guo-Ping Guo, Efficient Machine Learning Representations of Surface Code with Boundaries, Defects, Domain Walls and Twists, Phys. Rev. A 99, 012307 (2019)
- Yuan-Hang Zhang, **Z. Jia**, Yu-Chun Wu, and Guang-Can Guo, An Efficient Algorithmic Way to Construct Boltzmann Machine Representations for Arbitrary Stabilizer Code, [arXiv:1809.08631]
- **Z. Jia**, Rui Zhai, Bai-Chu Yu, Yu-Chun Wu, and Guang-Can Guo, Entropic No-Disturbance as a Physical Principle, Phys. Rev. A 97, 052128 (2018)
- Shang Yu, Chang-Jiang Huang, Jian-Shun Tang, **Z. Jia**, Yi-Tao Wang, Zhi-Jin Ke, Wei Liu, Zong-Quan Zhou, Ze-Di Cheng, Jin-Shi Xu, Yu-Chun Wu, Yuan-Yuan Zhao, Guo-Yong

Xiang, Chuan-Feng Li, Guang-Can Guo, Gael Sentís, and Ramon Muñoz-Tapia, Experimentally Detecting a Quantum Change Point via Bayesian Inference, *Phys. Rev. A* 98, 040301(R) (2018)

- Bai-Chu Yu, **Z. Jia**, Yu-Chun Wu, and Guang-Can Guo, Geometric Local Hidden State Model for Some Two-qubit States, *Phys. Rev. A* 98, 052345 (2018)
- Bai-Chu Yu, **Z. Jia**, Yu-Chun Wu, and Guang-Can Guo, Geometric Steering Criterion for Two-qubit States, *Phys. Rev. A* 97, 012130 (2018)
- **Z. Jia**, Gao-Di Cai, Yu-Chun Wu, Guang-Can Guo, and Adán Cabello, The Exclusivity Principle Determines the Correlation Monogamy, [arXiv:1707.03250]
- **Z. Jia**, Yu-Chun Wu, and Guang-Can Guo, Characterizing nonlocal correlations via universal uncertainty relations, *Phys. Rev. A* 96, 032122(2017)
- **Z. Jia**, Yu-Chun Wu, and Guang-Can Guo, Monogamy Relation in No-disturbance Theories, *Phys. Rev. A* 94, 012111(2016)
- Yan Shao, Fang-Ping Ouyang, Sheng-Lin Peng, Qi Liu, **Z. Jia**, Hui Zou, First-Principles Calculations of Electronic Properties of Defective Armchair MoS₂ Nanoribbons, [J]. *Acta Phys. -Chim. Sin.*, 2015,31 (11): 2083-2090.

LECTURE NOTES

- **Z. Jia**, Lecture notes on string theory
- **Z. Jia**, Lecture notes on quantum information theory